Algorithm 1 PPCV Framework

Input: LLM; original question q_0; number of paraphrased questions N; number of sampled alternative tokens K; temperature-scaling coefficient λ.
Output: Final answer ans_f.
(Paraphrasing) Generate paraphrased variants $q_1, q_2, \ldots, q_N$ from q_0 using the LLM.
(Initial reasoning) Obtain the initial reasoning path $r_0^{q_0}$ for q_0.
(Token influence analysis) For each paraphrased question q_n, compute the token distribution P^{q_n} by concatenating q_n with $r_0^{q_0}$ (Eq. 1).
(Candidate selection) Identify candidate critical token positions based on the mismatch between the top-1 predicted tokens and the expected tokens in q_0 (Eq. 2).
(Critical token verification) Select the final critical token a_c using the criteria (Eqs. 3–5).
(Alternative token sampling) Sample K alternative tokens $a_c^0, a_c^1, \ldots, a_c^{K-1}$ (including $a_c^0 = a_c$) using top-K sampling on q_0.
(Truncated rollouts) Truncate the initial reasoning path $r_0^{q_0}$ at position a_c and form synthetic inputs $\tilde{r}_c^0, \tilde{r}_c^1, \ldots, \tilde{r}_c^{K-1}$ by appending each alternative token.
for each synthetic input $\tilde{r}_c^k$, $k = 0, \ldots, K-1$ **do**
  Generate rollouts $r_k^{q_0}, r_k^{q_1}, \ldots, r_k^{q_N}$ for the original and paraphrased questions using the LLM.
end for
(Consistency verification) Compute the final answer ans_f using (similarity-weighted) paraphrase consistency (Eq. 6).

high-quality paraphrased questions.

C. Algorithm

The complete algorithm of our proposed PPCV framework is illustrated in Algo. 1.

D. Discussion

We have shown that replacing critical tokens can correct intermediate reasoning paths and lead to the correct answer. In this section, we examine how our method influences reasoning paths that are already correct. First, we conduct a case study on an instance with a correct answer, where we follow our pipeline to identify the critical token and roll out new reasoning paths using alternative tokens. As illustrated in Figure 9, the new rollouts also yield the correct answer. Second, our pipeline incorporates both the initial reasoning path $r_0^{q_0}$ and its paraphrased variants $r_0^{q_n}$ for evaluation. The robustness of correct rollouts across paraphrased forms ensures high consistency scores, allowing them to stand out as the final answer. These findings suggest that our pipeline preserves the performance of LLMs on problems that can already be solved correctly by CoT (Wei et al., 2022).

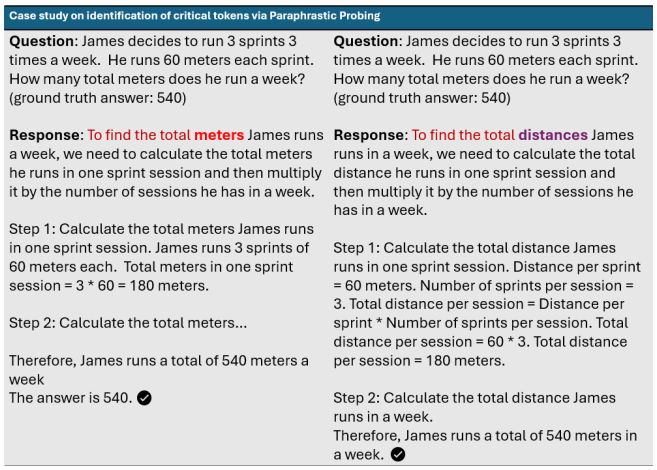


Figure 9. Effect of the critical token on an initial reasoning path that yields the correct answer. Critical tokens are highlighted in bold red, and alternative tokens in bold purple.

E. Main results

To further assess the effectiveness of our method, we evaluate it on a larger reasoning model such as DeepSeek-R1-Distill-Llama-70B. We apply our full pipeline and compare against all baselines across multiple mathematical reasoning benchmarks. The results, presented in Table 5, show that our method consistently outperforms the baselines. These improvements demonstrate that our approach remains effective for stronger reasoning models and generalizes well beyond the smaller models.

Table 5. Comparison between our method and baselines with DeepSeek-R1-Distill-Llama-70B model on mathematical reasoning benchmarks using pass@1 accuracy.

Method	AIME2024	AIME2025	BRUMO2025	HMMT2025
Chain-of-Thought	56.67	38.00	43.33	30.00
Predictive Decoding	60.00	40.66	44.66	30.66
Phi-Decoding	64.00	46.67	48.00	31.33
PPCV (Ours)	**70.00**	**56.66**	**56.66**	**33.33**

Table 6. Comparison of our method and the baseline approach on Llama-3.1-8B-Instruct, evaluated using pass@k (k=4).

Method	GSM8K	GSM-Hard	Math500	SVAMP	ARC
Phi-Decoding	92.15	53.57	52.60	91.19	90.73
PPCV (Ours)	**93.83**	**61.41**	**59.39**	**94.48**	**94.24**

Table 7. Comparison of our method and the baseline approach on Qwen3-32B, evaluated using pass@k (k=4).

Method	AIME2024	AIME2025	BRUMO2025	HMMT2025
Phi-Decoding	41.61	30.19	45.09	13.11
PPCV (Ours)	**49.71**	**34.28**	**51.42**	**19.28**

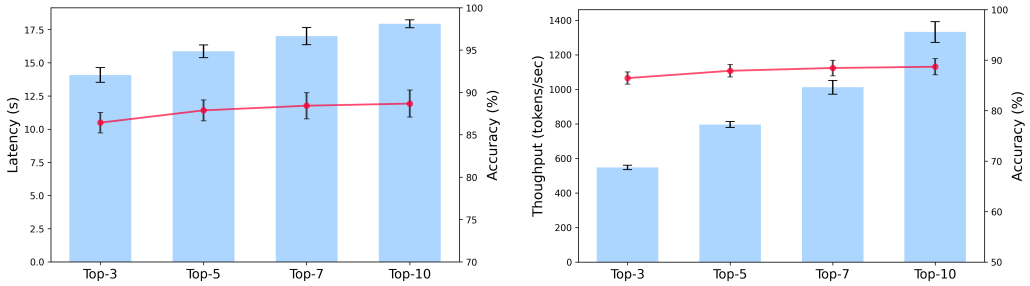
Table 8. The comparison of performance between Paraphrased Majority Voting (PMV) and our proposed PPCV using pass@1 accuracy.

Method	GSM8K	GSM-Hard	Math500	SVAMP	ARC
Chain-of-Thought	77.40	28.00	31.00	83.00	58.91
PMV	83.55	33.16	40.60	87.70	61.63
PPCV (Ours)	**88.24**	**49.73**	**50.00**	**89.60**	**88.31**

we also conduct additional experiments to report pass@k performance for both our method and the baselines. In these experiments, we use Phi-Decoding as the representative baseline and evaluate on two models: Llama-3.1-8B-Instruct and Qwen3-32B. We set k= 4 and generate 12 samples per question to obtain stable estimates of pass@4. The results across multiple benchmarks are presented in Table 6 and Table 7, respectively. Consistent with our main findings, our method achieves higher pass@k scores compared to the baseline methods, indicating that paraphrastic critical-token refinement continues to provide benefits in a multi-sample setting. These results further validate the robustness of our approach under stochastic sampling and demonstrate improved hit rates when multiple outputs are available.

F. Ablation Study

We also perform a comparison against a Paraphrased Majority Voting (PMV) strategy is essential to demonstrate that the performance gains of our method are indeed derived from the critical token correction, rather than simply the benefit of having multiple paraphrased samples. Thus, we have performed the suggested additional experiment. We adopt Paraphrased Majority Voting (PMV) as the baseline. For a fair comparison, this approach uses the same total number of trajectories generated from the paraphrased questions as PPCV, and reaches the final answer using standard majority voting over the rollouts from the paraphrased questions. Specifically, we use four variants of each question, with 10 rollouts per variant. We compare the PMV baseline against our full PPCV method and ensure the same number of paraphrased questions as in our method. The results shown in Table 8 demonstrate that the performance yielded by PMV is significantly lower than that of our proposed PPCV method. This result confirms that simply rolling out trajectories from paraphrased questions and aggregating the answers is insufficient. The substantial performance gap between PMV and PPCV decisively highlights the necessity and effectiveness of our extracted critical tokens and the subsequent corrective reasoning step.



(a) *(b)*

Figure 10. Trade-off between performance and efficiency for our method under different numbers of top-k tokens (k = 3, 5, 7, 10), measured by latency (left) and throughput (right).

G. Computational Cost Analysis

We also conduct a trade-off analysis between performance and efficiency by varying the number of alternative tokens considered at the critical position. Specifically, we test top-3, top-5, top-7, and top-10 alternatives using GSM8K, sampling 200 random questions and repeating the experiment five times to compute confidence intervals. The results are presented in Figure 10. We observe a clear trade-off. Reducing the number of alternative tokens lowers both latency and throughput, while causing a slight reduction in accuracy. This provides a practical mechanism for adjusting performance-efficiency trade-offs in real deployments. Depending on resource availability and target accuracy, practitioners can choose the appropriate number of alternative tokens.